



Notes on Bayesian Inference

Probabilistic Reasoning & Maximum Likelihood

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The Coin Flip Problem



Observe 10 flips: H H T T H H H H H T

7 heads, 3 tails

Key Question

What is $P(X = H)$? And what value of θ makes our observed data most likely?

Setup

Define $P(X = H) = \theta$, where $\theta \in [0,1]$. Each coin flip is a sample $X \sim P(X)$ from a Bernoulli distribution.

From Flips to Likelihood

The Binomial Distribution

The probability of k successes in n independent Bernoulli trials:

$$P(X=k \mid \theta) = C(n, k) \theta^k (1-\theta)^{n-k}$$

The Likelihood Function

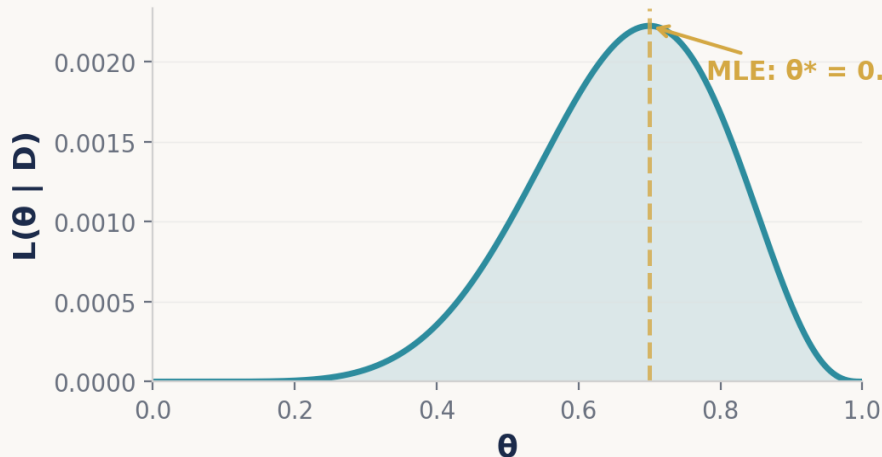
Same formula, viewed as a function of θ :

$$\mathcal{L}(\theta \mid D) = P(D \mid \theta)$$



$\mathcal{L}(\theta \mid D)$ and $P(D \mid \theta)$ are the same—but $\mathcal{L}$ emphasizes θ as the variable.

Likelihood Function: $\theta^7(1-\theta)^3$



For our data (7H, 3T): the curve peaks at $\vartheta = 0.7$

Maximum Likelihood Estimation

Find the θ that maximizes the likelihood of the observed data.

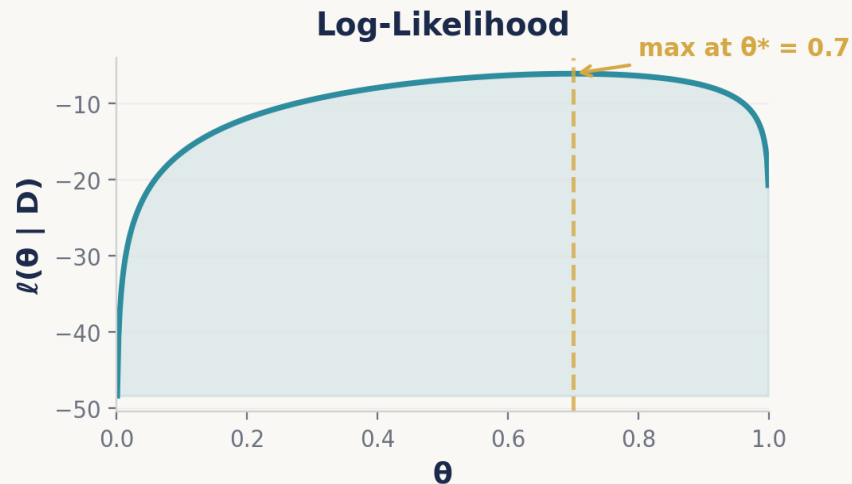
- 1 Take the log (monotonic $\rightarrow$ same max)

$$\ell(\theta) = k \log \theta + (n-k) \log(1-\theta)$$

- 2 Differentiate & set to zero

$$\partial \ell / \partial \theta = k/\theta - (n-k)/(1-\theta) = 0$$

$$\theta^*_{MLE} = k / n = 7 / 10$$



The log-likelihood is easier to differentiate—and has the same maximum.

Why MLE Isn't Enough

5 flips, 4 heads

$$\theta = 4/5 = 0.80$$

10 flips, 9 heads

$$\theta = 9/10 = 0.90$$

Does this feel right?

A real 1-franc coin probably isn't biased to 0.9. If 100 people flipped fair coins 10 times, roughly 1 would get 9 heads by chance.

Frequentist

There is a true, fixed parameter. Only data can inform estimates. More data $\rightarrow$ convergence.

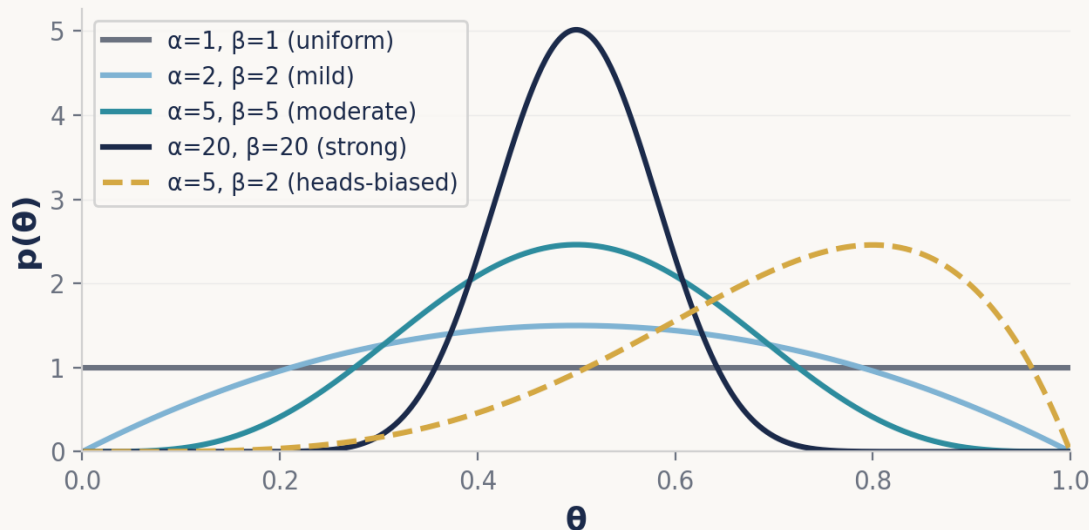
Bayesian

Incorporate prior beliefs before seeing data. Data updates those beliefs. Uncertainty is modeled explicitly.

The Prior: Beta Distribution

$$p(\theta) = [1 / B(\alpha, \beta)] \times \theta^{(\alpha-1)} \times (1-\theta)^{(\beta-1)}$$

Beta Prior Distributions



α and β

Shape parameters—think of them as pseudo-counts of prior heads (α) and tails (β).

Strength of belief

Larger $\alpha+\beta$ = more concentrated density = stronger prior conviction.

Your choice

The prior is subjective—a personal encoding of what you believe before seeing data.

Bayes' Rule & the Posterior

Updating beliefs after observing data

$$p(\theta \mid D) = p(D \mid \theta) \times p(\theta) / p(D)$$

posterior $\propto$ likelihood $\times$ prior

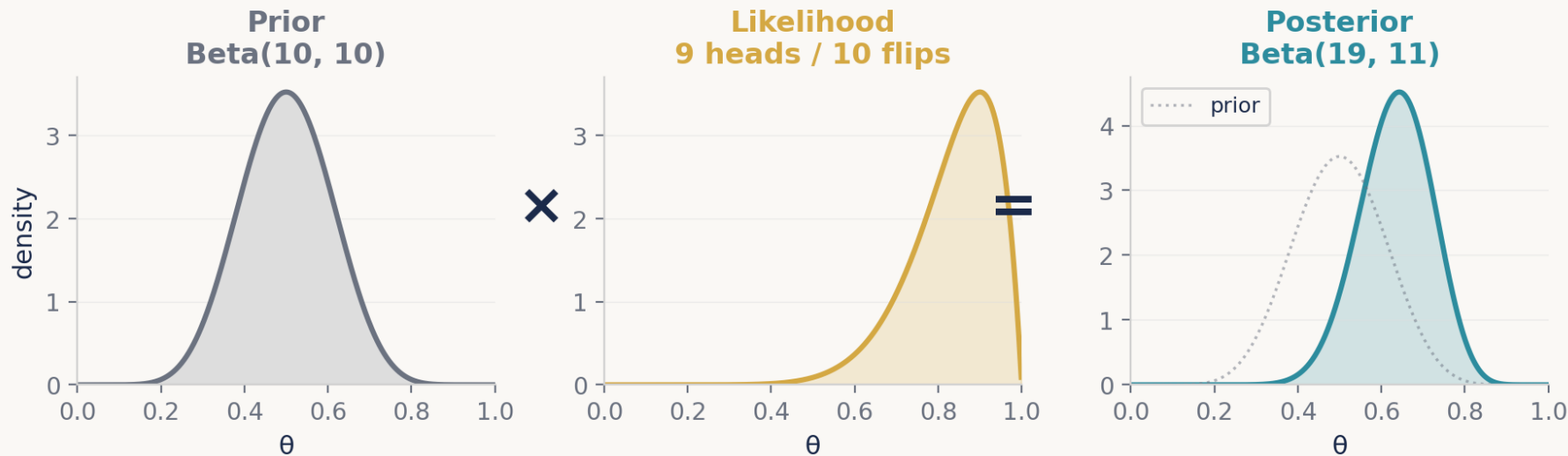
The denominator $p(D)$ is intractable but constant w.r.t. θ , so we drop it.

$$\theta^k (1-\theta)^{n-k} \times \theta^{\alpha-1} (1-\theta)^{\beta-1} = \theta^{k+\alpha-1} (1-\theta)^{n-k+\beta-1}$$

This has the form of another Beta distribution! $\rightarrow$

Prior $\times$ Likelihood = Posterior

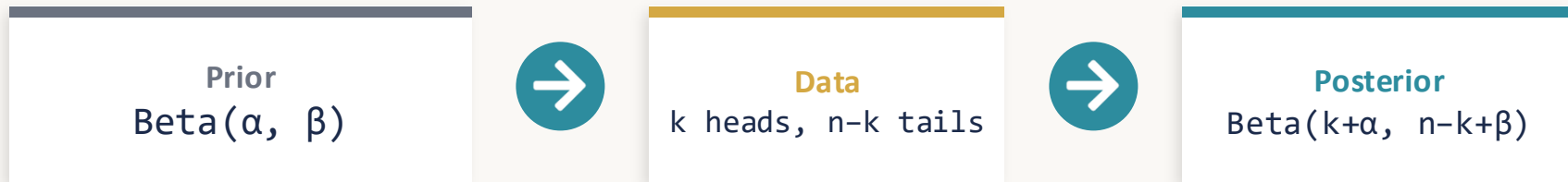
Example: Beta(10,10) prior with 9 heads in 10 flips



The posterior is a compromise: the strong fair-coin prior pulls the estimate back from the MLE of 0.9 toward 0.5.

Conjugate Prior: Closed-Form Posterior

The Beta prior + Binomial likelihood yields a Beta posterior.



$$p(\theta|D) = [1 / B(k+\alpha, n-k+\beta)] \times \theta^{(k+\alpha-1)} (1-\theta)^{(n-k+\beta-1)}$$

Interpretation: α and β act as pseudo-counts

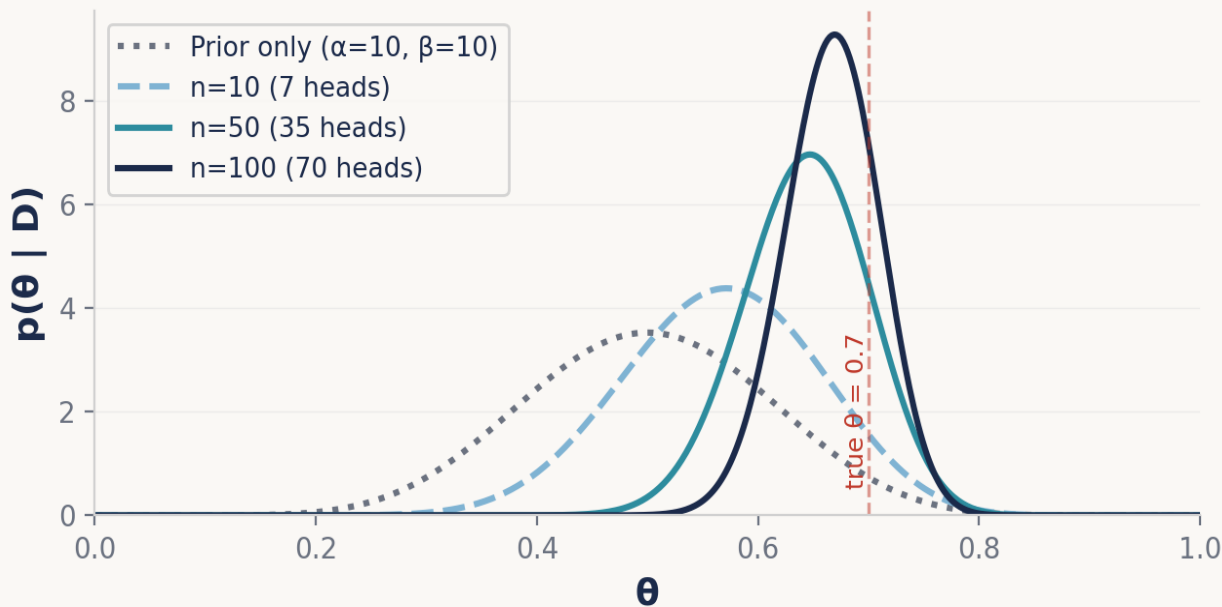
$$\theta^*_{\text{MAP}} = (\# \text{ observed heads} + \# \text{ prior heads}) / (\# \text{ observed flips} + \# \text{ prior flips})$$

As $n \rightarrow \infty$, the posterior concentrates around the true ϑ —the data overwhelms the prior.

Data Overwhelms the Prior

As we observe more data, the posterior concentrates around the true parameter value.

Posterior Concentrates with More Data



Small n

Prior has major influence.
Posterior is a compromise
between belief and evidence.

Large n

Data dominates. MAP and
posterior mean both
converge to the MLE: k/n .

MAP Estimation

Maximum A Posteriori: the mode of the posterior distribution.

$$\theta^*_{\text{MAP}} = (k + \alpha - 1) / (n + \alpha + \beta - 2)$$

	Formula	Notes
MLE	k / n	Data only; no prior
MAP	$(k + \alpha - 1) / (n + \alpha + \beta - 2)$	Mode of posterior
Post. Mean	$(k + \alpha) / (n + \alpha + \beta)$	Mean of posterior



MAP is the mode, not the mean. When the posterior is symmetric they coincide, but in general they differ. You don't need the normalizing constant to compute the MAP—just maximize the unnormalized posterior.

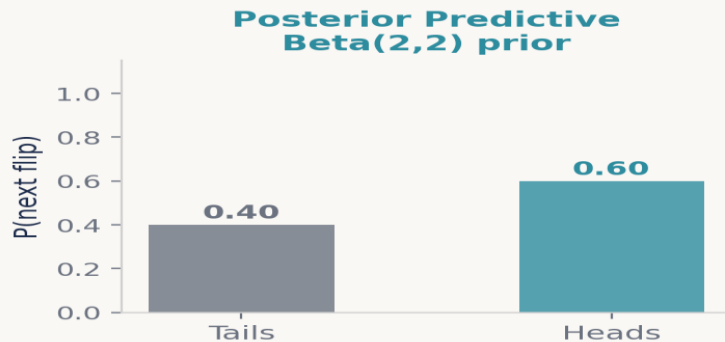
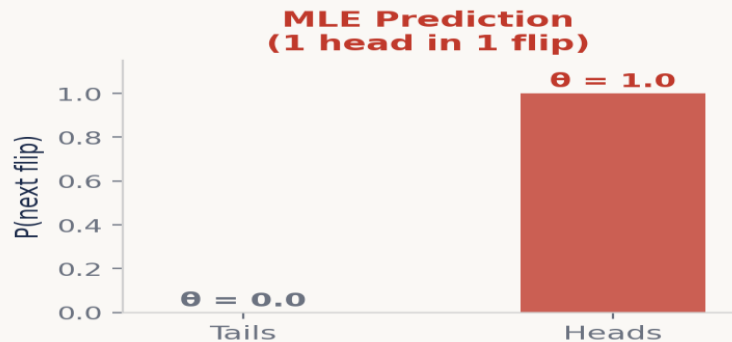
The Posterior Predictive

What is the probability of heads on the next flip?

Don't plug in a point estimate—average over all θ , weighted by the posterior.

$$P(X=H \mid D) = \int \theta \cdot p(\theta|D) d\theta = E[\theta|D] = (k+\alpha) / (n+\alpha+\beta)$$

Example: 1 flip, 1 head, Beta(2,2) prior



Summary

- 1 Likelihood** Measures how probable the observed data is for a given θ .
 - 2 MLE** Maximizes the likelihood. For the Binomial: $\theta^* = k/n$.
 - 3 Prior** Encodes beliefs about θ before seeing data. Beta(α, β) for coin flips.
 - 4 Posterior** Updated beliefs after data. Posterior $\propto$ likelihood $\times$ prior.
 - 5 MAP** Mode of the posterior: $(k + \alpha - 1) / (n + \alpha + \beta - 2)$.
 - 6 Predictive** Averages predictions over all θ , weighted by the posterior.
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